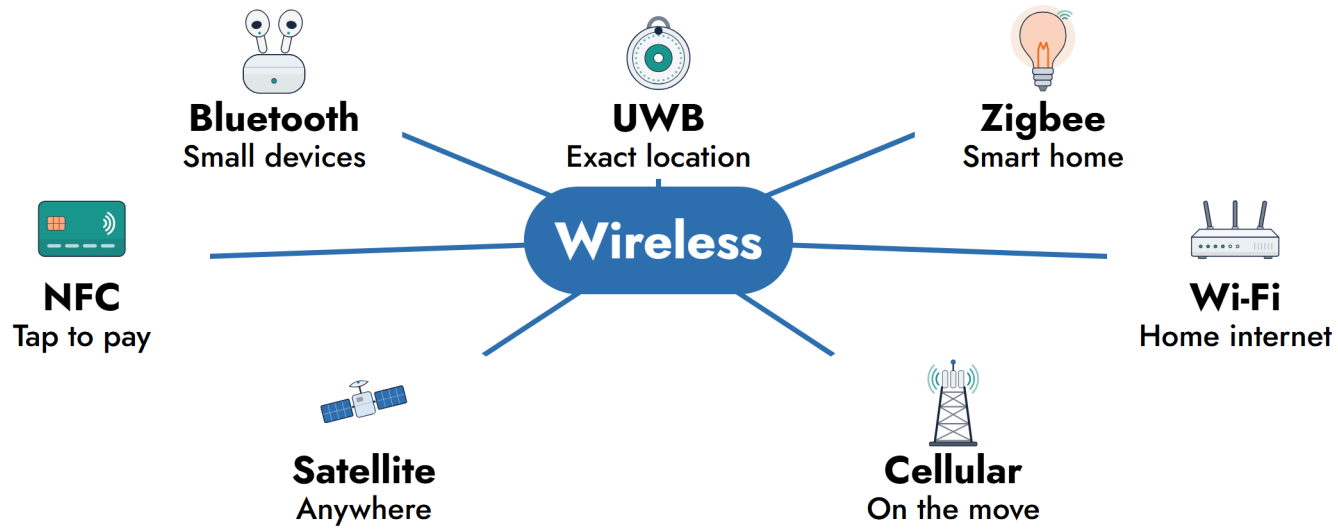
# EVERY WIRELESS TECHNOLOGY

The cheat sheet, from a few centimetres to outer space




## Side by side

| Technology | Typical range | Speed | Power use | Best for |
|---|---|---|---|---|
| **NFC** | About 4 cm | Up to 424 kbps | None in card | Payments, key cards |
| **Bluetooth** | About 10 m | Up to 3 Mbps | Low | Earbuds, watches |
| **UWB** | Tens of metres | Short messages | Low | Trackers, car keys |
| **Zigbee** | 10–20 m per hop | Up to 250 kbps | Very low | Bulbs, sensors |
| **Wi-Fi** | About 30 m indoors | 100s of Mbps | Medium | Homes, offices |
| **Cellular** | Up to km per cell | 10s–100s of Mbps | Medium | Phones, cars |
| **Satellite** | Most of the planet | 50–100 Mbps | High | Ships, remote areas |

## How each one works

**NFC:** The reader's magnetic field powers the card's chip, and the chip replies by changing how much energy it takes.

**Bluetooth:** Classic Bluetooth hops between channels over 1,000 times a second, so interference only costs a tiny piece.

**UWB:** Times very short pulses to measure distance, usually to about 10 cm. Several antennas give the direction.

**Zigbee:** Plugged-in devices relay messages in a mesh, while battery devices sleep and wake only to send.

**Wi-Fi:** The radio link between your device and the router. It is not the internet. Devices take turns to send.

**Cellular:** Your phone connects to a nearby tower, and the network hands you to the next tower as you move.

**Satellite:** Signals go up to a satellite and down to a ground station. Higher orbits cover more but add delay.

## Which one is it?

Pay with a tap
**NFC**

Listen on earbuds
**Bluetooth**

Find your lost keys
**UWB**

Run battery sensors for a year
**Zigbee**

Stream video at home
**Wi-Fi**

Stay online on a bus
**Cellular**

Connect with no towers nearby
**Satellite**

Typical values. Real results vary with the device, the band and the surroundings. Power use is a rough comparison.

**Watch the full video: Every Wireless Technology Explained, Prime Explained, youtube.com/@theprimexplained**